

Database Management Systems: Assignment-5

Name: Suraj Shah

Admission Number: I24AI012

Q1. Use Employee table from Assignment 2.

- a) Display the list of employees whose name starts with 'A'.

Query:

```
SELECT * FROM employee WHERE Name LIKE 'A%';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
123	Ajay Singh	IT	203	2018-09-05	95000
124	Anjali Verma	HR	201	2020-02-01	68000

- b) Add a new column named city and display the list of employees whose city starts with letter 'm'.

Query:

```
ALTER TABLE employee ADD City varchar(25);
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary	City
121	Ravi Kumar	HR	201	2019-03-15	75000	Mumbai
122	Sneha Sharma	Finance	202	2021-06-10	82000	Delhi
123	Ajay Singh	IT	203	2018-09-05	95000	New York
124	Anjali Verma	HR	201	2020-02-01	68000	Pune
125	Rahul Gupta	Marketing	204	2022-01-20	55000	Toronto
126	Siddharth Jain	IT	203	2017-11-12	105000	San Francisco
127	Shivani Desai	Finance	202	2020-08-15	72000	London
128	Sunil Kumar	HR	201	2023-04-12	60000	Ahmedabad
129	Kavita Mehta	Operations	205	2019-12-18	50000	Sydney
130	Pooja Patel	IT	203	2021-09-25	88000	Chennai

Query:

```
SELECT * FROM employee WHERE City LIKE 'm%';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary	City
121	Ravi Kumar	HR	201	2019-03-15	75000	Mumbai

- c) Display the list of employees whose city ends with word 'pradesh'.

Query:

```
SELECT * FROM employee WHERE City LIKE '%pradesh';
```

Output:

Empty set (0.001 sec)

- d) Display the list of employees who are having exactly five characters in their name.

Query:

```
SELECT * FROM employee WHERE Name LIKE '_____ %';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary	City
122	Sneha Sharma	Finance	202	2021-06-10	82000	Delhi
125	Rahul Gupta	Marketing	204	2022-01-20	55000	Toronto
128	Sunil Kumar	HR	201	2023-04-12	60000	Ahmedabad
130	Pooja Patel	IT	203	2021-09-25	88000	Chennai

- e) Display the list of employees whose city is having only five characters.

Query:

```
SELECT * FROM employee WHERE City LIKE '_____';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary	City
122	Sneha Sharma	Finance	202	2021-06-10	82000	Delhi

- f) Find the names of employees whose name includes the substring "man".

Query:

```
SELECT * FROM employee WHERE Name LIKE '%man%';
```

Output:

```
Empty set (0.001 sec)
```

Q2. Use Student table from Assignment 2.

- a) Display the list of students whose name starts with 'N'.

Query:

```
SELECT * FROM student WHERE Name LIKE 'N%';
```

Output:

Empty set (0.001 sec)

- b) Add a new column named branch and display the list of employees whose branch starts with letter 'A'

Query:

```
ALTER TABLE student ADD Branch varchar(30);
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom	Branch	City
101	Suraj Shah	4	2001-05-14	2019-07-01	201	Artificial Intelligence	Mumbai
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305	Information Tech	Delhi
103	Ravi Patel	4	2000-11-10	2018-07-01	202	Electronics	Ahmedabad
104	Sneha Sharma	2	2003-01-25	2021-07-01	105	Mechanical	Pune
105	Ajay Singh	4	2001-08-05	2019-07-01	402	Civil	Chennai
106	Sunil Kumar	1	2004-06-18	2022-07-01	210	Electrical	Jaipur
107	Anjali Verma	3	2002-12-30	2020-07-01	115	Chemical	Surat
108	Siddharth Jain	4	2001-03-12	2019-07-01	203	Computer Science	Lucknow
109	Rahul Gupta	2	2003-09-05	2021-07-01	312	Civil	Hyderabad
110	Shivani Desai	4	2000-10-28	2018-07-01	205	Mechanical	Indore

Query:

```
SELECT * FROM student WHERE Branch LIKE 'A%';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom	Branch
101	Suraj Shah	4	2001-05-14	2019-07-01	201	Artificial Intelligence

- c) Display the list of students who are having exactly five characters in their name.

Query:

```
SELECT * FROM student WHERE Name LIKE '____ %';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom	Branch	City
101	Suraj Shah	4	2001-05-14	2019-07-01	201	Artificial Intelligence	Mumbai
104	Sneha Sharma	2	2003-01-25	2021-07-01	105	Mechanical	Pune
106	Sunil Kumar	1	2004-06-18	2022-07-01	210	Electrical	Jaipur
109	Rahul Gupta	2	2003-09-05	2021-07-01	312	Civil	Hyderabad

- d) Add a new column named city and display the list of students whose city is having only ten characters.

Query:

```
ALTER TABLE student ADD City varchar(25);
```

```
SELECT * FROM student where City like '_____';
```

Output:

Empty set (0.001 sec)

- e) Find the names of students whose name includes the substring “sh”.

Query:

```
SELECT * FROM student WHERE Name LIKE '%sh%';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom	Branch	City
101	Suraj Shah	4	2001-05-14	2019-07-01	201	Artificial Intelligence	Mumbai
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305	Information Tech	Delhi
104	Sneha Sharma	2	2003-01-25	2021-07-01	105	Mechanical	Pune
110	Shivani Desai	4	2000-10-28	2018-07-01	205	Mechanical	Indore

Q3. Create table Customers with this schema: (CustomerID, CustomerName, ContactName, Address, City, PostalCode, Country)

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Rameshbhai & Co	Ramesh Patel	123 MG Road	Ahmedabad	380001	India
2	Satnam Traders	Satish Kumar	45 Ring Road	Lucknow	226001	India
3	Global Corp	Aman Singh	12 Main Street	Madhya Pradesh	462001	India
4	Sureshat Sales	Suresh Bhatt	5 Hill Street	Mumbai	400001	India
5	Sanjay Exporters	Sanjay Mehta	88 Park Lane	Surat	395001	India

- a) Return all customers that ends with the pattern 'bhai'.

Query:

```
SELECT * FROM customers WHERE CustomerName LIKE '%bhai';
```

Output:

Empty set (0.001 sec)

- b) Return all customers that contains the pattern 'sat'.

Query:

```
SELECT * FROM customers WHERE CustomerName LIKE '%sat%';
```

Output:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Satnam Traders	Satish Kumar	45 Ring Road	Lucknow	226001	India

- c) Return all customers with a city starting with any character, followed by "pradesh".

Query:

```
SELECT * FROM customers WHERE City LIKE '%pradesh';
```

Output:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
3	Global Corp	Aman Singh	12 Main Street	Madhya Pradesh	462001	India

- d) Return all customers with a city starting with "S", followed by any four characters, ending with "at".

Query:

```
SELECT * FROM customers WHERE City LIKE 'S____at';
```

Output:

Empty set (0.001 sec)

- e) Return all customers from Mumbai.

Query:

```
SELECT * FROM customers WHERE City='Mumbai';
```

Output:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
4	Sureshat Sales	Suresh Bhatt	5 Hill Street	Mumbai	400001	India